

SQL Metadata Guide

This document provides business metadata used by the ESG Natural Language to SQL Generator. It describes analytical entities, dimensions, metrics, aggregation rules, business synonyms, and SQL generation guidance. It is intended to be indexed in Chroma and retrieved during SQL generation.

Primary Dataset

All analytical queries target the table `esg_data`. Only read-only SELECT statements are supported.

Business Entities

Facility, Country, City, Industry and Fiscal Year are business entities used for grouping, filtering and drill-down.

Dimensions

Facility_Name, Country, City, Industry and Fiscal_Year are dimensions. Use dimensions in GROUP BY clauses. Multiple dimensions may be combined when explicitly requested.

Metrics

Scope1_Emissions, Scope2_Emissions, Scope3_Emissions, Total_Energy_MWh, Renewable_Energy_Percentage, Water-Withdrawal_m3, Water_Recycled_Percentage, Waste_Generated_Tons, Waste_Recycled_Percentage, Employee_Count, Female_Workforce_Percentage, Training_Hours and Compliance_Violations are analytical metrics.

Aggregation Rules

SUM → Emissions, Energy, Water Withdrawal, Waste Generated, Employee Count, Compliance Violations.
AVG → Renewable Energy %, Water Recycled %, Waste Recycled %, Female Workforce %, Training Hours.
COUNT → Number of Facilities or Records.

Sorting Rules

Highest / Top → ORDER BY metric DESC.
Lowest / Bottom → ORDER BY metric ASC.
Rankings should always include LIMIT when the user specifies Top N or Bottom N.

Trend Analysis

Trend questions should group by Fiscal_Year and sort Fiscal_Year ascending.

Comparison Analysis

Comparison questions should GROUP BY the requested business dimension such as Country, Industry or Facility.

Drill-down Hierarchy

Global → Country → City → Industry → Facility.
Preserve existing filters unless the user explicitly replaces them.

Business Synonyms

Carbon Footprint → Scope1 + Scope2 + Scope3 Emissions.

Carbon Emissions → Scope Emissions.

Green Energy → Renewable_Energy_Percentage.

Women Representation / Gender Diversity → Female_Workforce_Percentage.

Headcount / Workforce → Employee_Count. Plant / Site → Facility_Name.

Filtering Guidance

Country filters apply to Country.

Industry filters apply to Industry.

Fiscal year references apply to Fiscal_Year. Facility names should filter Facility_Name.

Visualization Mapping

Trend → Line Chart.

Comparison → Bar Chart.

Ranking → Horizontal Bar Chart. Distribution / Share → Pie Chart.

Single KPI → Metric Card.

Correlation → Scatter Plot.

Unsupported Operations

Never generate INSERT, UPDATE, DELETE, DROP, ALTER, CREATE, MERGE, TRUNCATE, PRAGMA, ATTACH or DETACH. Only SELECT statements are permitted.

Executive Reporting Guidance

Executive summaries should explain business findings, highlight significant comparisons, identify highest and lowest values where meaningful, discuss observable trends and avoid SQL terminology.